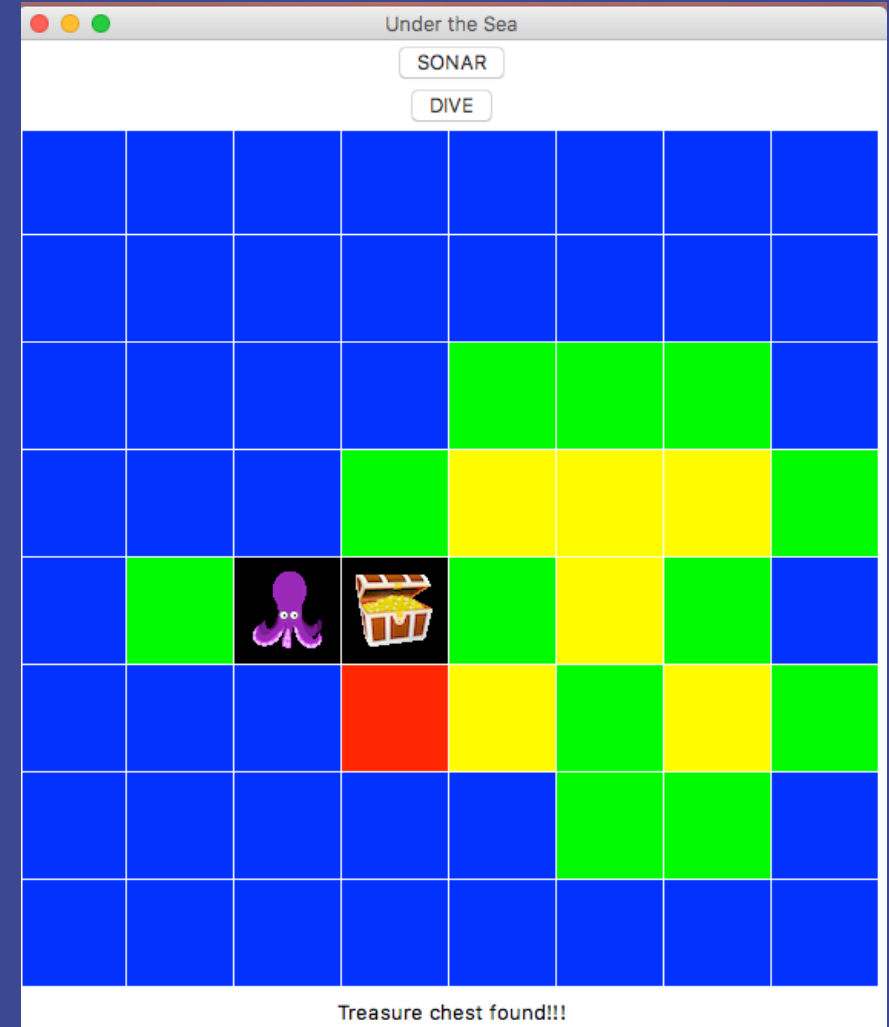


CS 156

Intro to AI

Uncertainty



Some slides are based on the slides created by Dan Klein and Pieter
Abbeel for CS188 Intro to AI at UC Berkeley.
The artwork is by Ketrina Yim

Where we are

So far we have talked about:

- ▶ Problem Solving / Search
- ▶ Reasoning / Planning

Coming up : the tools that got us out of the AI winter

- ▶ Uncertainty
- ▶ Learning

Where we are

1970-90: Knowledge-based approaches

- ▶ 1969-79: Early development of knowledge-based systems
- ▶ 1980-88: Expert systems industry booms
- ▶ 1988-93: Expert systems industry busts: "AI Winter"

1990-: Statistical approaches

- ▶ Resurgence of probability, focus on uncertainty
- ▶ General increase in technical depth
- ▶ Agents and learning systems... "AI Spring"?

Uncertainty?

Uncertainty comes into play when the environment is:

- ▶ Partially observable. Partial observability may be due to:
 - Incomplete knowledge
 - Faulty sensors/limited sensors
- ▶ Non-deterministic:
 - actions do not always go as planned – swimming in the ocean (waves and rip currents)
 - Faulty actuators

Probabilistic Reasoning

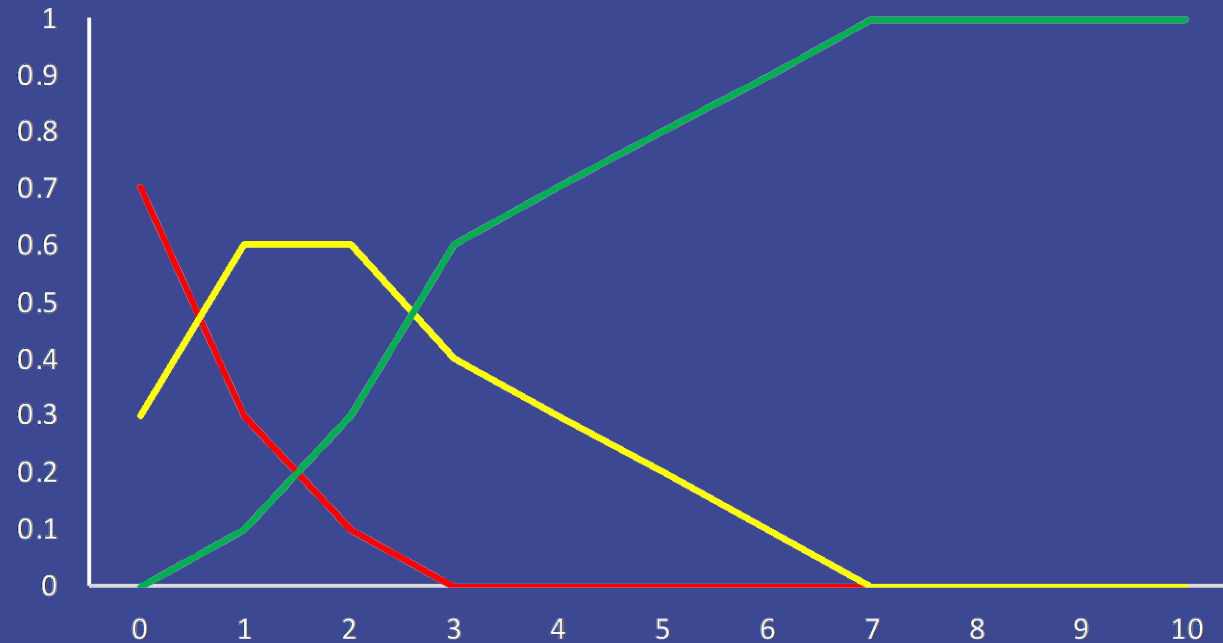
A framework for managing our **beliefs** and knowledge

- ▶ Observed variables (**evidence**): Agent knows certain things about the state of the world (sensor readings or symptoms)
- ▶ **Unobserved** variables: Agent needs to reason about other aspects (where an object is or what disease is present)
- ▶ **Model**: Agent knows something about how the known variables relate to the unknown variables.

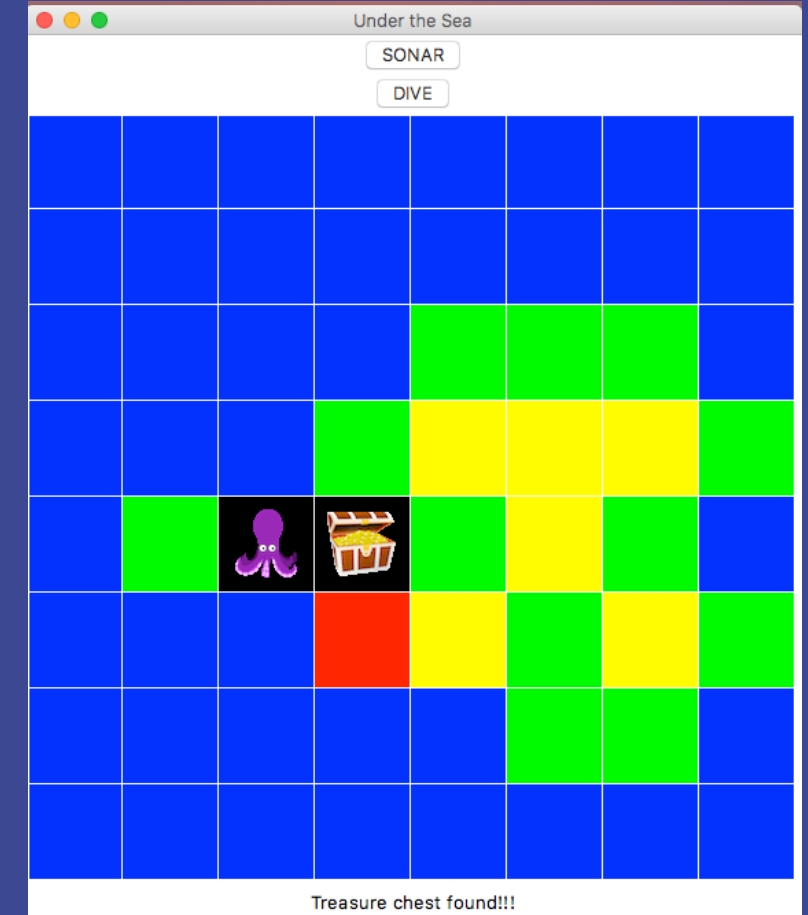
Agent can also **learn the model** (coming up in a couple of weeks)

Probabilistic Reasoning Example

Color
Probability

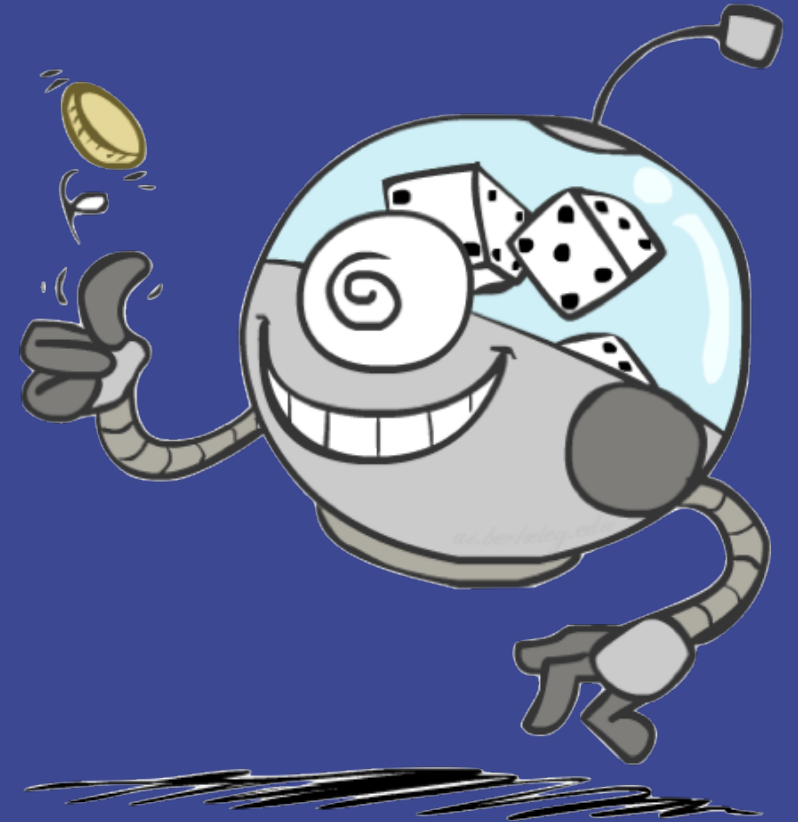


Manhattan distance
from treasure



Random Variables

- ▶ A random variable represents some aspect of the world about which we (may) have uncertainty
 - R: Is it raining?
 - T: hot or cold?
 - D: How long will it take to drive to work?
 - L: Where is the treasure?
- ▶ We denote random variables with capital letters
- ▶ Like variables in a CSP, random variables have domains
 - R in {true, false} (often write as {+r, -r})
 - T in {hot, cold}
 - L in possible locations, maybe {(0,0), (0,1), ...}



Probability Distributions

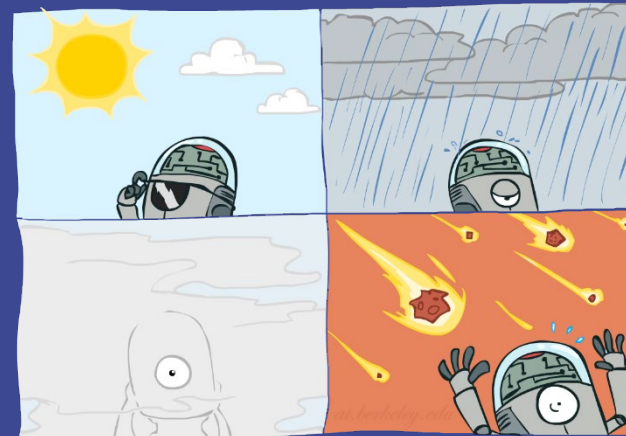
- ▶ Associate a probability with each value in the domain
 - Temperature: hot or cold?
 - Weather: sun, rain, fog, ...?

$P(T)$



T	P
hot	0.5
cold	0.5

$P(W)$



W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Distributions

- **Unobserved** random variables have distributions

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

- A distribution is a TABLE of probabilities of values
- A probability is a single number: $P(W = \text{rain}) = 0.1$
- We must have:
 - $\forall x P(X = x) \geq 0$
 - $\sum_x P(X = x) = 1$

Shorthand Notation:

- $P(\text{hot}) = P(T=\text{hot})$
- $P(\text{cold}) = P(T=\text{cold})$
- ...

OK *if* all domains are disjoint

iClicker

We have not taken any sonar reading yet.

What is the probability that the treasure is in the top left square?



Uniform Probability Distribution

$P(L)$

L	P
(0, 0)	0.04
(0, 1)	0.04
(0, 2)	0.04
(0, 3)	0.04
(0, 4)	0.04
(1, 0)	0.04
...	0.04
...	0.04
(4, 4)	0.04



Joint Distribution

A joint distribution is a probability distribution that involves two or more random variables.

Random variable
T: hot or cold?



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Random variable
W: sun or rain?



For n variables with domain size d , the size of the joint distribution is d^n .

Probabilistic Models

- ▶ A probabilistic model is defined by a joint distribution over a set of random variables
 - (Random) variables with domains
 - Assignments are called **outcomes**
 - Joint distributions: how likely assignments (outcomes) are
 - **Normalized**: sum to 1.0
 - Ideally: only certain variables directly interact

Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Events

- ▶ An **event** is a set E of outcomes
- ▶ From a joint distribution, we can calculate the probability of any event
 - Probability that it's hot AND sunny?
 - Probability that it's hot?
 - Probability that it's hot OR sunny?
- ▶ Typically, the events we care about are **partial assignments**, like $P(T=\text{hot})$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Marginal Distributions

- ▶ **Marginal** distributions are sub-tables which eliminate some variables
- ▶ Marginalization (summing out): Combine collapsed rows by adding

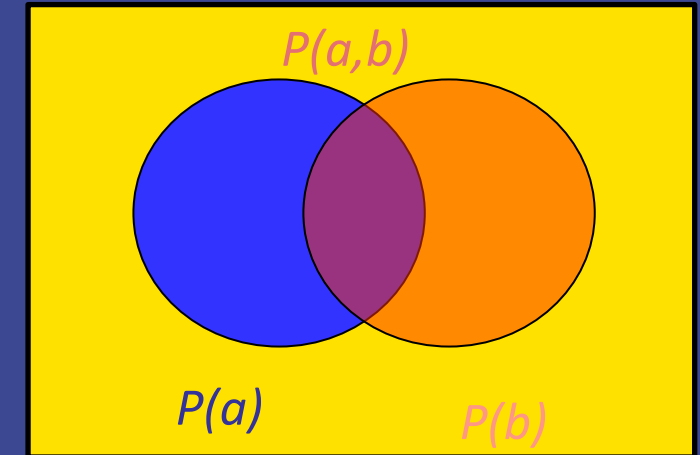
$P(T, W)$				$P(T)$	
T	W	P		T	P
hot	sun	0.4	→	hot	0.5
hot	rain	0.1		cold	0.5
cold	sun	0.2		$P(W)$	
cold	rain	0.3	→	W	P
				sun	0.6
				rain	0.4

Conditional Probabilities

- Definition of conditional probability

Given two random variables A and B:

$$P(a|b) = \frac{P(a, b)}{P(b)}$$



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

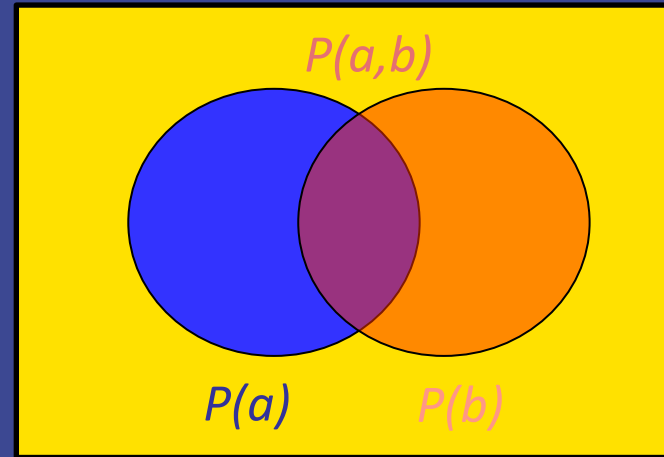
$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$$\begin{aligned} P(T=c) &= P(T=c, W=s) + P(T=c, W=r) \\ &= 0.2 + 0.3 = 0.5 \end{aligned}$$

iClicker: Conditional Probabilities

- Definition of conditional probability:

$$P(a|b) = \frac{P(a,b)}{P(b)}$$



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(r|h) = ?$$

Conditional Probabilities

- Definition of conditional probability

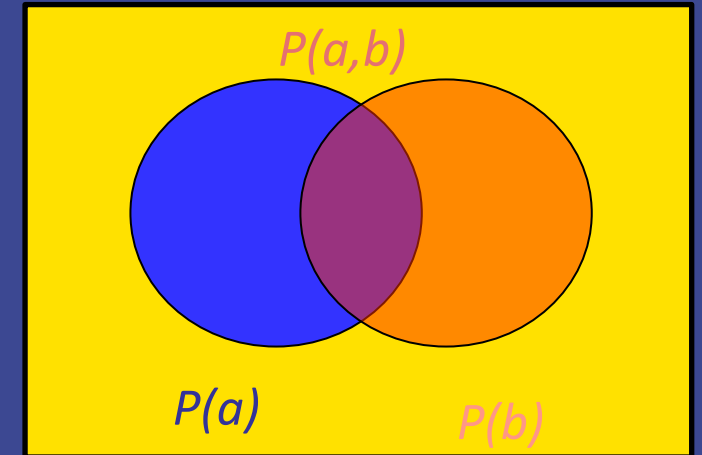
Given two random variables A and B:

$$P(a|b) = \frac{P(a,b)}{P(b)}$$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(r|h) = \frac{P(r, h)}{P(h)} = \frac{0.1}{0.5} = 0.2$$



Conditional Distributions

- Conditional distributions are probability distributions over some variables **given fixed values of others**

Conditional Distributions

$P(W|T)$

W	P
sun	0.8
rain	0.2

W	P
sun	0.4
rain	0.6

Joint Distribution

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Computing Conditional Distributions

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\begin{aligned}P(W=s|T=c) &= \frac{P(W=s, T=c)}{P(T=c)} \\&= \frac{P(W=s, T=c)}{P(W=s, T=c) + P(W=r, T=c)} \\&= \frac{0.2}{0.5} = 0.4\end{aligned}$$

$P(W|T=c)$

W	P
sun	0.4
rain	0.6


$$\begin{aligned}P(W=r|T=c) &= \frac{P(W=r, T=c)}{P(T=c)} \\&= \frac{P(W=r, T=c)}{P(W=s, T=c) + P(W=r, T=c)} \\&= \frac{0.3}{0.5} = 0.6\end{aligned}$$

Normalization Trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence



$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)



$P(W|T=c)$

W	P
sun	0.4
rain	0.6

Step 1: Compute $Z = \text{sum over all entries } (0.2 + 0.3)$
 Step 2: Divide every entry by Z

$$\begin{aligned}
 P(W=s|T=c) &= \frac{P(W=s, T=c)}{P(T=c)} \\
 &= \frac{P(W=s, T=c)}{P(T=c, W=s) + P(T=c, W=r)} \\
 &= \frac{0.2}{0.5} = 0.4
 \end{aligned}$$

$$\begin{aligned}
 P(W=r|T=c) &= \frac{P(W=r, T=c)}{P(T=c)} \\
 &= \frac{P(W=r, T=c)}{P(T=c, W=s) + P(T=c, W=r)} \\
 &= \frac{0.3}{0.5} = 0.6
 \end{aligned}$$

The Product Rule

- Sometimes we have conditional distributions but want the joint distribution:

$$P(x|y) = \frac{P(x, y)}{P(y)} \quad \longleftrightarrow \quad P(y)P(x|y) = P(x, y)$$

$P(y)$ is a marginal distribution.
It is also called the prior distribution

The Chain Rule

- ▶ More generally, we can always write any joint distribution as an incremental product of conditional distributions:

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1) P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i|x_1 \dots x_{i-1})$$

Bayes' Rule

Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y) P(y) = P(y|x) P(x)$$

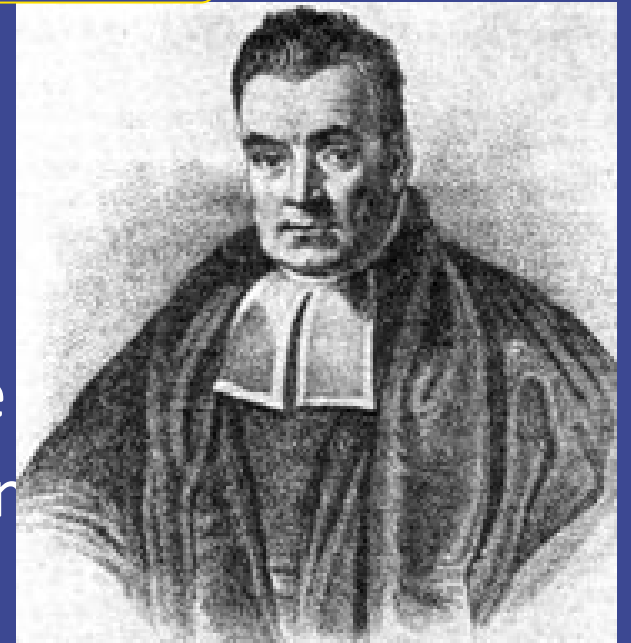
Dividing by $P(y)$, we get: $P(x|y) = \frac{P(y|x)}{P(y)} P(x)$

Why is this at all helpful?

- ▶ Lets us build one conditional from its reverse
- ▶ Often one conditional is tricky but the other is simple
- ▶ Foundation of many systems in AI (speech recognition, machine translation)

In the running for most important AI equation!

That's my rule!



Probabilistic Inference

Probabilistic inference: compute a desired probability from other known probabilities

- ▶ We generally compute conditional probabilities
 - These represent the agent's *beliefs* given the *evidence*
- ▶ Probabilities change with new evidence:
 - Observing *new evidence causes beliefs to be updated*

Inference with Bayes' Rule

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

Example: A robot needs to infer location from (noisy) sensor reading

- ▶ Cause: Location
- ▶ Effect: Sensor reading
- ▶ Easy to measure $P(\text{Sensor} | \text{Location})$
- ▶ We compute $P(\text{Location} | \text{Sensor})$

Inference with Bayes' Rule

Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

Example: M: meningitis, S: stiff neck

- ▶ $P(+m) = 0.0001$
 - ▶ $P(+s | +m) = 0.8$
 - ▶ $P(+s | -m) = 0.01$
- } givens

A patient has a stiff neck. What is the probability that they have meningitis?

$P(+m | +s) =$

Inference with Bayes' Rule

Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

Example: M: meningitis, S: stiff neck

- ▶ $P(+m) = 0.0001$
 - ▶ $P(+s|+m) = 0.8$
 - ▶ $P(+s|-m) = 0.01$
- } gives

$$P(+s) = P(+s, +m) + P(+s, -m)$$

$$\text{Product rule: } P(y)P(x|y) = P(x, y)$$

$$P(+s, +m) = P(+m) P(+s|+m)$$

$$P(+s, -m) = P(-m) P(+s|-m)$$

$$P(+s) = P(+m) P(+s|+m) + P(-m) P(+s|-m)$$

A patient has a stiff neck. What is the probability that they have meningitis?

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)}$$

What is $P(+s)$?

Inference with

Diagnostic probability from causal probability

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

Example: M: meningitis, S: stiff neck

- ▶ $P(+m) = 0.0001$
 - ▶ $P(+s|+m) = 0.8$
 - ▶ $P(+s|-m) = 0.01$
- } givens

Someone has a stiff neck. What is the probability that they have meningitis?

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)}$$

What is $P(+s)$?

iClicker:

What is $P(-m)$?

$$P(+s) = P(+s, +m) + P(+s, -m)$$

$$\text{Product rule: } P(y)P(x|y) = P(x, y)$$

$$P(+s, +m) = P(+m) P(+s|+m)$$

$$P(+s, -m) = P(-m) P(+s|-m)$$

$$P(+s) = P(+m) P(+s|+m) + P(-m) P(+s|-m)$$

Inference with Bayes' Rule

Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

Example: M: meningitis, S: stiff neck

- ▶ $P(+m) = 0.0001$
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- } givens

$$P(+s) = P(+s, +m) + P(+s, -m)$$

$$\text{Product rule: } P(y)P(x|y) = P(x, y)$$

$$P(+s, +m) = P(+m) P(+s|+m)$$

$$P(+s, -m) = P(-m) P(+s|-m)$$

$$P(+s) = P(+m) P(+s|+m) + P(-m) P(+s|-m)$$

Someone has a stiff neck. What is the probability that they have meningitis?

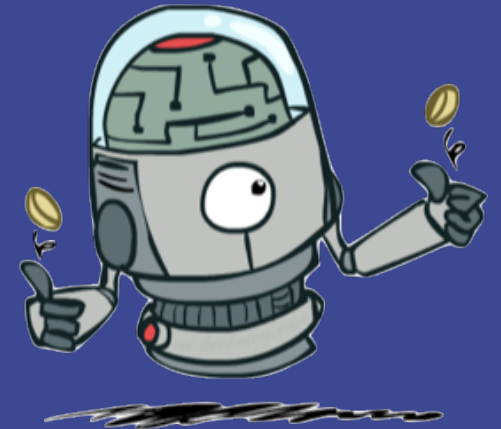
$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{0.8 \times 0.0001}{0.0001 \times 0.8 + 0.9999 \times 0.01} = 0.008$$

Independence

- ▶ Two random variables are **independent** in a joint distribution if:

$$P(X, Y) = P(X)P(Y)$$

$$\forall x, y, P(x, y) = P(x)P(y) \quad X \perp\!\!\!\perp Y$$



- ▶ the joint distribution **factors** into a product of two simple ones
- ▶ Usually variables aren't independent!
- ▶ Can use independence as a **modeling assumption**
- ▶ Independence can be **a simplifying assumption**

Independence?

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4

$P_i(T, W) = P(T) P(W)$

T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

No Independence!

Conditional Independence

- ▶ Unconditional (absolute) independence is very rare
- ▶ *Conditional independence* is our most basic and robust form of knowledge about uncertain environments
- ▶ X is conditionally independent of Y given Z $X \perp\!\!\!\perp Y \mid Z$

if and only if:

$$\forall x, y, z \ P(x, y|z) = P(x|z)P(y|z)$$

or equivalently if and only if:

$$\forall x, y, z \ P(x|z, y) = P(x|z)$$

Conditional Independence

When we write: $P(x, y | z)$

We mean: $P((x, y) | z)$

When we write: $P(x | z, y)$

We mean: $P(x | (z, y))$

Conditional Independence

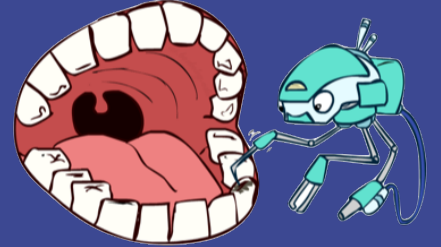
$P(\text{Toothache}, \text{Cavity}, \text{Catch})$

- ▶ If someone has a cavity, the probability that the probe gets caught doesn't depend on whether they have a toothache:

- $P(+\text{catch} \mid +\text{cavity}, +\text{toothache}) = P(+\text{catch} \mid +\text{cavity})$
- $P(+\text{catch} \mid +\text{cavity}, -\text{toothache}) = P(+\text{catch} \mid +\text{cavity})$

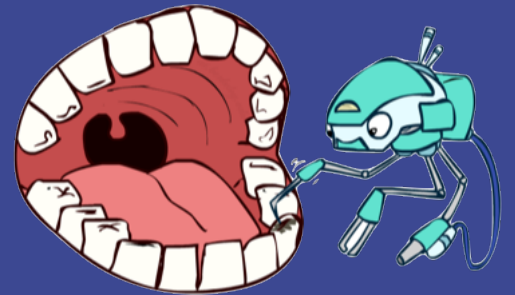
- ▶ The same independence holds if they don't have a cavity:

- $P(+\text{catch} \mid -\text{cavity}, +\text{toothache}) = P(+\text{catch} \mid -\text{cavity})$
- $P(+\text{catch} \mid -\text{cavity}, -\text{toothache}) = P(+\text{catch} \mid -\text{cavity})$



Conditional Independence

- ✓ $P(+catch \mid +cavity, +toothache) = P(+catch \mid +cavity)$
- ✓ $P(+catch \mid +cavity, -toothache) = P(+catch \mid +cavity)$
- ✓ $P(+catch \mid -cavity, +toothache) = P(+catch \mid -cavity)$
- ✓ $P(+catch \mid -cavity, -toothache) = P(+catch \mid -cavity)$
- ▶ $P(-catch \mid +cavity, +toothache) = P(-catch \mid +cavity)$
- ▶ $P(-catch \mid +cavity, -toothache) = P(-catch \mid +cavity)$
- ▶ $P(-catch \mid -cavity, +toothache) = P(-catch \mid -cavity)$
- ▶ $P(-catch \mid -cavity, -toothache) = P(-catch \mid -cavity)$



Catch is *conditionally independent* of Toothache given Cavity:

- ▶ $P(Catch \mid Cavity, Toothache) = P(Catch \mid Cavity)$

Conditional Independence

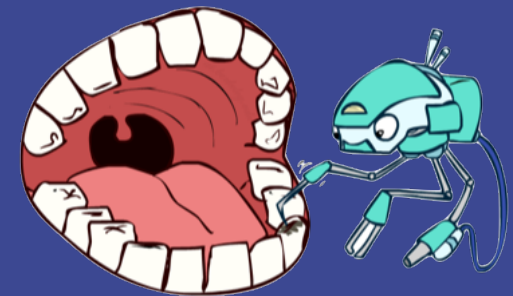
Catch is *conditionally independent* of Toothache given Cavity:

Equivalent statements:

$$P(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})$$

$$P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) P(\text{Catch} \mid \text{Cavity})$$

Toothache is conditionally independent of Catch given Cavity



Why?

- ▶ Bayes Nets and Probabilistic Reasoning
- ▶ ...
- ▶ Machine Learning: Naive Bayes Classification

Reminder

- ▶ Homework 5 due by 5 PM